

Implementation Summary – WBOCRM

Project: Web-Based Operational CRM

Course: Istanbul Arel University – HW5 Part 2

Date: May 2026

Completed Phases

Phase 1 – Project Setup

- Monorepo structure: `backend/` (Node.js) and `frontend/` (React/Vite)
- Git repository initialised, `.gitignore` configured
- `backend/package.json` with all production and dev dependencies
- `frontend/package.json` with React, React Router, Axios, Vite

Phase 2 – Database Layer

- PostgreSQL schema in `backend/src/db/schema.sql`
- Tables: `UserAccount`, `Lead`, `InteractionLog`, `SupportTicket`
- `pool.js` with SSL auto-detection for Render deployment
- `init.js` (schema creation) and `seed.js` (demo data with bcrypt-hashed passwords)

Phase 3 – Backend API

- 3-tier architecture: Controller → Service → Repository
- Auth: POST `/api/auth/login`, POST `/api/auth/register`
- Leads: full CRUD + GET sorted by score + CSV export
- Tickets: full CRUD with optimistic locking
- Interaction logs: GET + POST per lead
- Dashboard: active leads, open tickets, top 5 leads, monthly revenue
- Users: GET all, PUT role (admin only)

Phase 4 – Middleware & Security

- `authMiddleware.js` : verifies JWT, attaches `req.user`
- `rbacMiddleware.js` : `allowRoles(...roles)` factory — returns 403 if role insufficient
- Input validation via `express-validator` on auth routes
- All SQL uses parameterised queries

Phase 5 – Lead Scoring Algorithm

- `scoringService.js` : 5-metric weighted scoring (0–100)
- Metrics: calls (×25), meetings (×25), budget (×20), companySize (×20), emailOpens (×10)
- Score calculated at lead creation time and stored in DB

Phase 6 – Frontend

- `AuthContext` with JWT decode, login/logout, persisted in `localStorage`
- `api.js` Axios instance: auto-injects `Authorization: Bearer` header
- Protected routing: `ProtectedLayout` + `RoleRoute` in `App.jsx`
- Pages:
 - `LoginPage` — form with `user_email` / `user_password` fields
 - `DashboardPage` — 4 KPI cards with 2-second auto-refresh
 - `LeadPage` — table with create/edit/delete/profile navigation + CSV export button

- `LeadProfilePage` — unified customer profile: lead info + interaction timeline + linked tickets + admin assignment
- `TicketPage` — ticket list with create/update (optimistic locking conflict detection)
- `UsersPage` — admin-only user management with role dropdown
- Components: `NavBar` , `LeadForm` , `TicketForm` , `PriorityTable`
- Responsive layout, minimum viewport 360px

Phase 7 – Testing

- **18 unit tests** (Jest + mocked repositories): `authService`, `scoringService`, `ticketService`
- **6 functional tests** (Jest + Supertest, live DB): lead CRUD, ticket update, auth 401/403
- **11 BDD scenarios** (Cucumber): lead registration, ticket creation, auth flows, scoring
- All tests passing; results saved to `backend/test_results/`

Phase 8 – Deployment Configuration

- `render.yaml` : Blueprint defining backend + frontend + PostgreSQL services
- `backend/.env.example` and `frontend/.env.example`
- `deployment_info.txt` : step-by-step deployment guide

Phase 9 – Documentation

- `docs/architecture_description.md` : 3-tier diagram, design decisions, schema
- `docs/user_stories.md` : 14 user stories across 6 epics with READ references
- `docs/acceptance_tests.md` : BDD scenarios mapped to user stories and NFRs
- `docs/implementation_summary.md` : this file
- `README.md` : full technical documentation for public repository

Technology Stack

Layer	Technology	Version
Backend runtime	Node.js	20 LTS
Backend framework	Express	4.18
Database driver	pg (node-postgres)	8.11
Auth	jsonwebtoken	9.x
Password hashing	bcryptjs	2.4
Validation	express-validator	7.x
Frontend framework	React	18.x
Build tool	Vite	5.x
Routing	React Router	v6
HTTP client	Axios	1.x
Unit/functional tests	Jest + Supertest	29.x / 6.x
BDD tests	@cucumber/cucumber	10.x

Database	PostgreSQL	15 (Render)
Deployment	Render	—

READ Compliance

Section	Requirement	Status
FR-AUTH-01/02	JWT login / 401 on invalid	✓
FR-AUTH-03/04	401 no token / 403 wrong role	✓
FR-LM-01 – FR-LM-05	Lead CRUD	✓
FR-LM-06	Interaction log notes	✓
FR-SC-01	Priority score 0–100	✓
FR-SC-06	Lead assignment to sales rep	✓
FR-ST-01/02	Ticket create / update	✓
FR-DA-01	Dashboard KPIs	✓
FR-DOC-15	CSV export	✓
FR-UM-01/02	User listing / role update	✓
NFR-ST-05	Optimistic locking	✓
NFR-ST-12	Dashboard 2s auto-refresh	✓
Section 3.8.3	Unified Customer Profile	✓
Section 14	BDD acceptance tests	✓

Known Constraints

- Free Render tier: services spin down after 15 min inactivity (30s cold start)
- JWT expiry is 30 minutes; no refresh token mechanism (scope per READ)
- Scoring recalculated only at creation — editing metrics after creation does not update score (by design)
- Monthly revenue uses `created_at` as the date proxy for Closed leads